

Project Planning Phase

Project Title: Diabetic Retinopathy Detection Using Deep Learning

1. Introduction

Project planning is an important step in project development. It helps in organizing the tasks, defining project goals, and setting a timeline for completing the project successfully. In this phase, the overall structure and schedule of the project are prepared.

The main goal of this project is to develop a deep learning model that can detect diabetic retinopathy from retinal images.

2. Project Objectives

- The objectives of this project are:
- To design an automated system for detecting diabetic retinopathy.
- To use deep learning techniques for medical image analysis.
- To improve the accuracy and efficiency of disease detection.
- To assist healthcare professionals in early diagnosis.

3. Project Scope

The project focuses on developing a system that can analyze retinal images and detect diabetic retinopathy using deep learning techniques. The system will classify images and provide prediction results to assist doctors.

4. Project Milestones

Milestone	Description
Problem Identification	Understanding the project problem
Literature Survey	Studying existing research
Data Collection	Collecting retinal image datasets
Model Development	Designing and training the model
Testing	Evaluating model performance
Documentation	Preparing final project report

6. Tools and Technologies

- Programming Language: Python
- Deep Learning Framework: TensorFlow / Keras
- Image Processing: OpenCV
- Development Environment: Jupyter Notebook
- Version Control: GitHub

